Determinants of Household Participation in Securities Markets

An Empirical Analysis of the 2025 SEBI Household Survey

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The Problem

Only 25.5% of Indian households invest in securities.

Despite the post-2020 retail boom and zero-commission platforms, participation remains sharply stratified by socioeconomic position, geography, and information access.

Four research questions:

1. Who is likely to participate?
2. What keeps the 74.5% out?
3. Which securities do investors prefer, and for how long?
4. Can we predict these outcomes from basic profiles?

Sample

Total respondents	109,430
Active investors	27,882 (25.5%)
Non-investors	81,548 (74.5%)
Geographic zones	4
States / UTs covered	35

Methods

Logistic regression (HC3 robust SE, survey-weighted), K-means clustering, Histogram-based Gradient Boosting

Dataset and Schema: Details

Source: 2025 SEBI Household Survey, nationally representative, pre-computed survey weights.

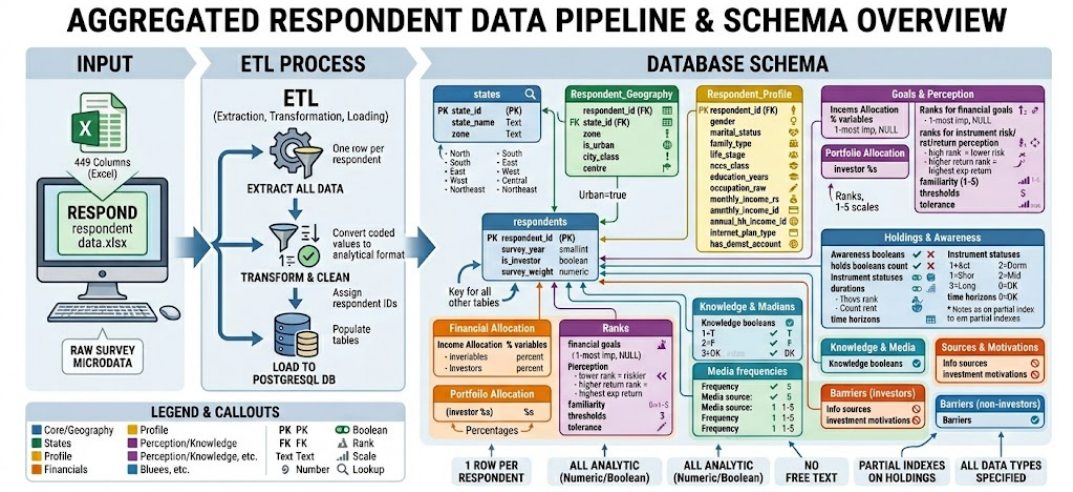
ETL pipeline: 449 raw columns transformed into boolean, integer, and ordinal types; loaded into a normalized PostgreSQL schema (17 domain-specific tables anchored by `respondent_id`).

Conditional routing:

- Investors: sub-module SS_B3 (info sources, motivations), $n = 9{,}797$
- Non-investors: sub-module SS_BB2 (barriers), $n = 14{,}794$

All group percentages computed against the routing-eligible population.

Dataset and Schema: Architecture



Methodology (I): Econometric Model

Multivariate Participation Logic

$$\begin{aligned}\text{logit}(p_i) = \beta_0 &+ \beta_1 \text{Awareness}_i + \beta_2 \ln(\text{Income}_i) + \beta_3 \text{Education}_i \\ &+ \beta_4 \text{Male}_i + \beta_5 \text{Urban}_i + \beta_6 \text{Risk}_i \\ &+ \sum_j \delta_j \text{Cohort}_{ij} + \sum_l \lambda_l \text{Zone}_{il} + \epsilon_i\end{aligned}$$

- **Dependent Variable:** p_i is the probability of market participation ($Y = 1$).
- **Survey Weighting:** All coefficients estimated using inverse probability weighting to match national demographics.
- **Robustness:** HC3 Huber-White robust standard errors to account for complex survey design and non-normality.

Methodology (II): Unsupervised & Predictive Layer

Barrier Archetypes (Finding 2)

- **Algorithm:** K-Means clustering.
- **Feature Space:** 18-dimensional binary vector of self-reported barriers.
- **Optimization:** Elbow method and Silhouette scores for cluster validation ($n = 3$ archetypes).

Predictive Modeling (Layer)

- **Model:** Histogram-based Gradient Boosting.
- **Capability:** Handles categorical features natively; captures complex non-linear interactions.
- **Validation:** 5-fold cross-validation; AUC-ROC as primary metric.

EDA: Who Participates?

Income Gradient

10.7% → 47.6%

Q1 (<Rs. 12,500) to Q5 (>Rs. 35,000): a **4.5**× spread. Participation nearly doubles with each income step.

Urban Premium

28.8% vs 17.6%

Urban households are **1.6**× more likely to participate. The gap widens further in lower income brackets.

Gender Gap

30.8% vs 15.8%

Male participation is nearly **2**× female. The gap persists across every income level and zone.

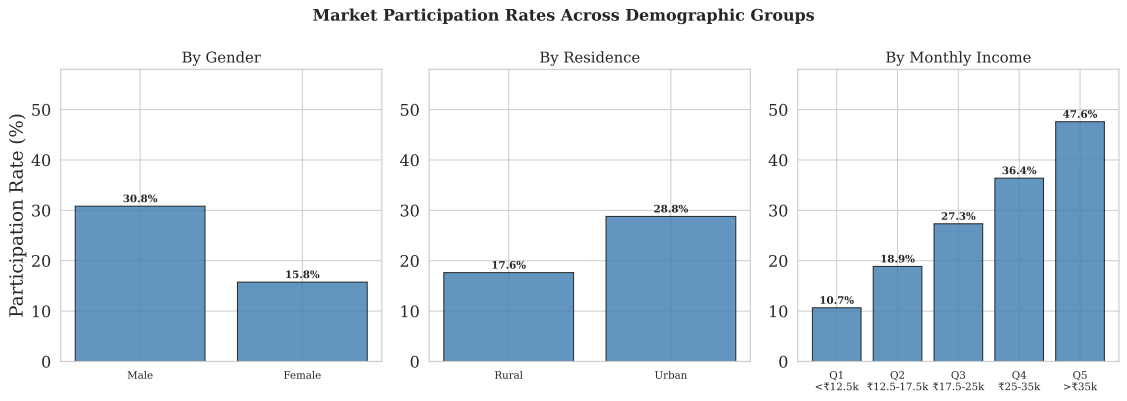
Generational Peak

Millennials: 28.1%

Ahead of Gen Z (24.3%) and Gen X (22.9%). Digital access appears to drive uptake more than experience.

All four effects survive joint multivariate controls → Finding 1.

EDA: Participation Demographics



EDA: Awareness Gap and Holdings

Awareness (all 109,430 respondents):

Fixed Deposits	**99.1%**
Mutual Funds/ETFs	67.0%
Direct Equity	62.0%
Derivatives	17.9%

Top holdings among investors:

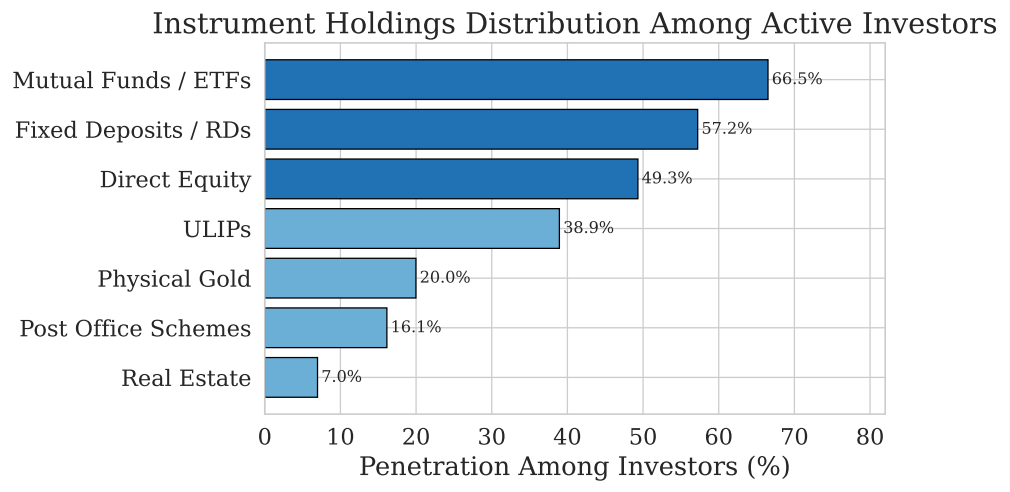
Mutual Funds/ETFs	66.5%
Fixed Deposits	57.2%
Direct Equity	49.3%

MFs and FDs co-dominate: investors layer market products on top of savings rather than replacing them.

The Awareness-Participation Gap

67% of households know mutual funds exist, yet only 25.5% invest in any security. Awareness is necessary but not sufficient: trust, knowledge depth, and access remain barriers.

EDA: Instrument Awareness and Holdings



Finding 1: What Drives Market Entry?

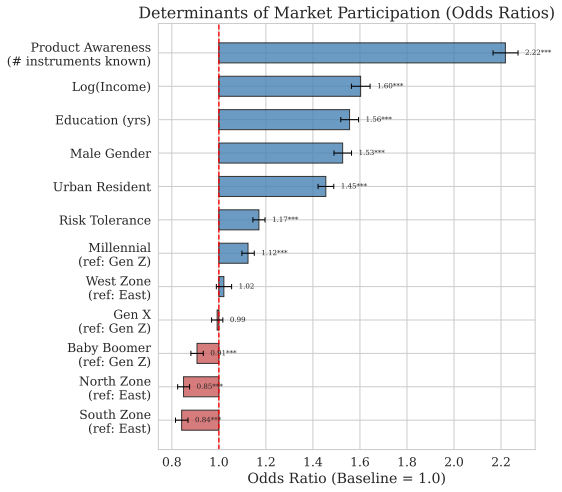
Product awareness is the single strongest predictor (OR = 2.22), exceeding:

Income (log)	OR = 1.60
Education	OR = 1.56
Male gender	OR = 1.53
Urban	OR = 1.45

Partial mediation: income and education partly work *through* awareness, but awareness retains a large independent effect after SES controls.

Two predictors return null results (Gen X, West zone): the model does not produce spurious significance at $N \approx 100{,}000$.

Finding 1: Odds Ratios for Participation



Finding 2: Three Archetypes of Non-Participation

K-means clustering on 18 self-reported barriers identifies three distinct profiles:

Trust-Deficient (46.3%)

Urban, higher-income households who *could* invest but choose not to. Top barriers: lack of trust in institutions and regulatory concerns. Requires accountability measures: grievance redressal, broker transparency.

Knowledge-Gated (29.0%)

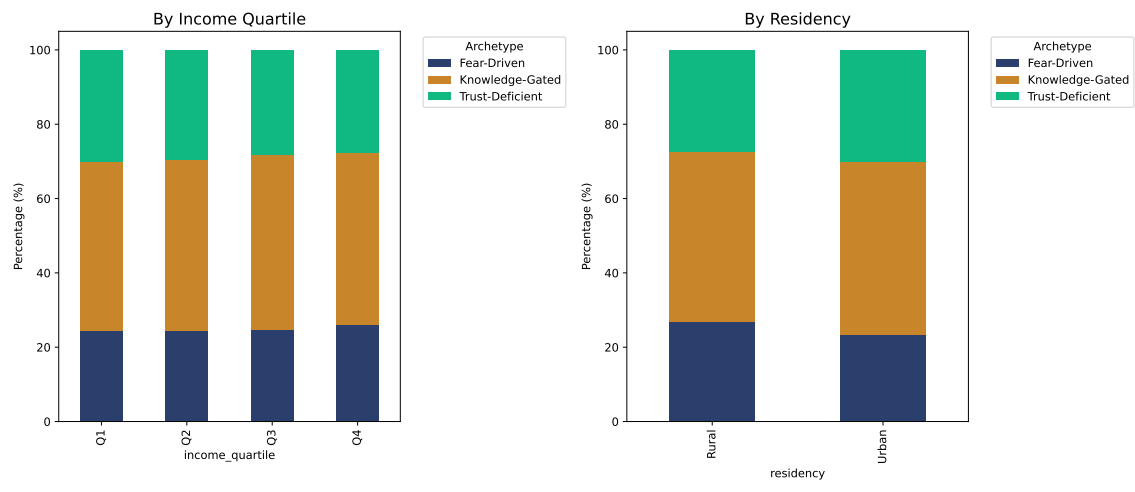
Rural, lowest income quartile. Top barrier: "I don't know how to start." Needs geographically targeted literacy programs in local languages.

Fear-Driven (24.8%)

Spread across all demographics. Top barrier: fear of losing money. Responds to low-risk entry points that reduce perceived downside.

A single awareness campaign cannot address all three.

Finding 2: Archetype Demographics



Finding 3: Behavioral Traps Inside the Market

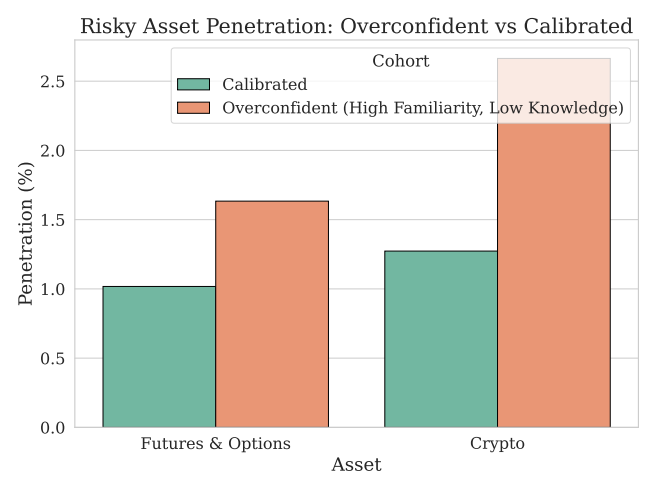
Dunning-Kruger effect: Investors with high perceived familiarity but low objective knowledge hold F&O and crypto at significantly higher rates than calibrated peers.

Finfluencer pipeline: 5.0% of investors cite social media as primary information source; this cohort is disproportionately quick-gains oriented relative to professionally-advised investors.

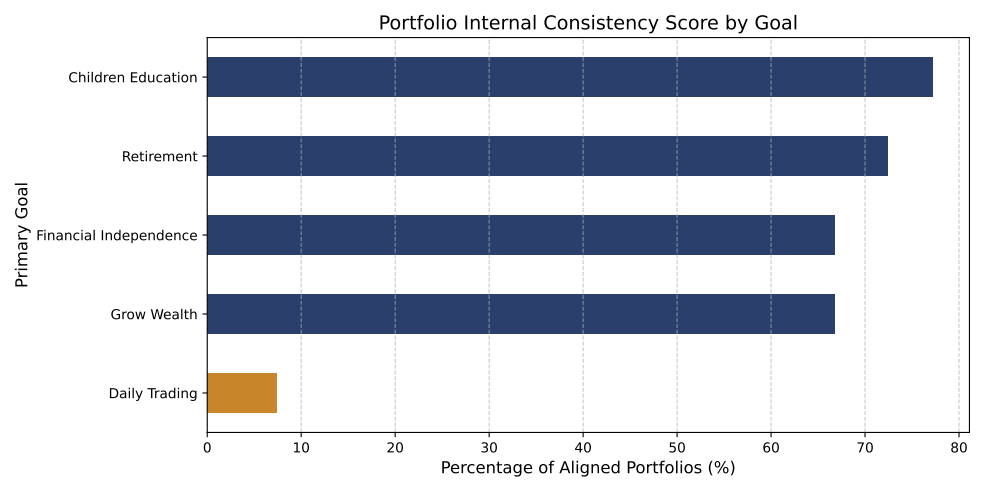
Goal-portfolio gap: The "Daily Trading" cohort has a goal-portfolio consistency score of 7.3%: they identify as speculators without holding the instruments that operationalize that strategy.

These phenomena compound: aspirational investors, shaped by finfluencer content, are the segment most exposed to adverse outcomes.

Finding 3: Dunning-Kruger and Holdings



Finding 3: Goal-Portfolio Consistency



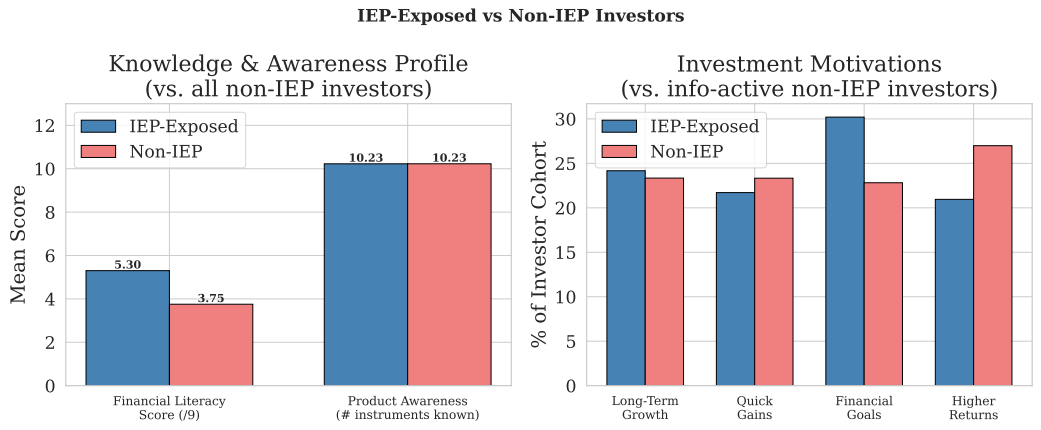
Finding 4: Investor Education Programme Impact

Only 2,768 investors (9.9%) cite SEBI or industry IEP as a top information source. Yet the difference in behaviour is striking:

Metric	IEP	Non-IEP
Knowledge score (out of 9)	**5.30**	3.75
Products aware	10.23	10.23
Goal-based motivation	**30.2%**	22.8%
Quick-gains motivation	21.7%	23.3%
Holds Fixed Deposits	49.3%	**55.1%**

The 41% knowledge advantage is in *depth*, not breadth: both groups are aware of the same number of products. IEP also reshapes *how* investors participate: quick-gains motivation falls while goal-based motivation rises, a shift from speculative to intentional investing. The lower FD rate reflects this: educated investors substitute FDs for market-linked products. Current reach: 9.9%.

Finding 4: IEP Analysis



Predictive Layer: Machine Learning Results

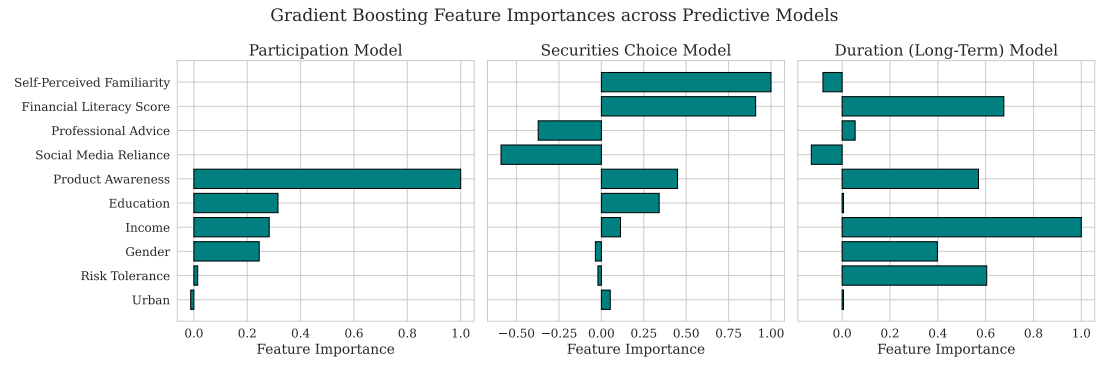
Model	Task	AUC
M1	Participation	**0.814**
M2	Asset choice	**0.820**
M3	Horizon	0.612

Feature importance shift:
At the entry layer, *product awareness* dominates. Inside the market, *perceived familiarity* and *objective knowledge* take over.

Investment horizon (M3) is the hardest prediction, consistent with it being driven by situational factors outside a cross-sectional feature space.

All three models include `n_products_aware` alongside demographic and behavioral features.

Predictive Layer: Feature Importance



**Note: Negative permutation importance indicates high-variance predictors (e.g., social media usage) that may appear noisy when shuffled individually. However, HistGradientBoosting utilizes these features for critical conditional interactions; their removal breaks high-precision decision branches, leading to lower aggregate AUC.*

Policy Recommendations

A. Prioritise Product Awareness

Awareness predicts participation more than income. Campaigns should mention MFs, ETFs, SIPs by name.

B. Tailor outreach by non-investor type

Trust-Deficient: institutional accountability.
Knowledge-Gated: vernacular rural delivery.
Fear-Driven: low-risk entry points.

C. Expand investor education

SEBI's programs reach only 9.9% of investors. Scale via workplaces and Jan Dhan accounts.

D. Close the gender gap

Women invest at half the rate of men even after controlling for income. Women-only groups and lower minimum amounts would help.

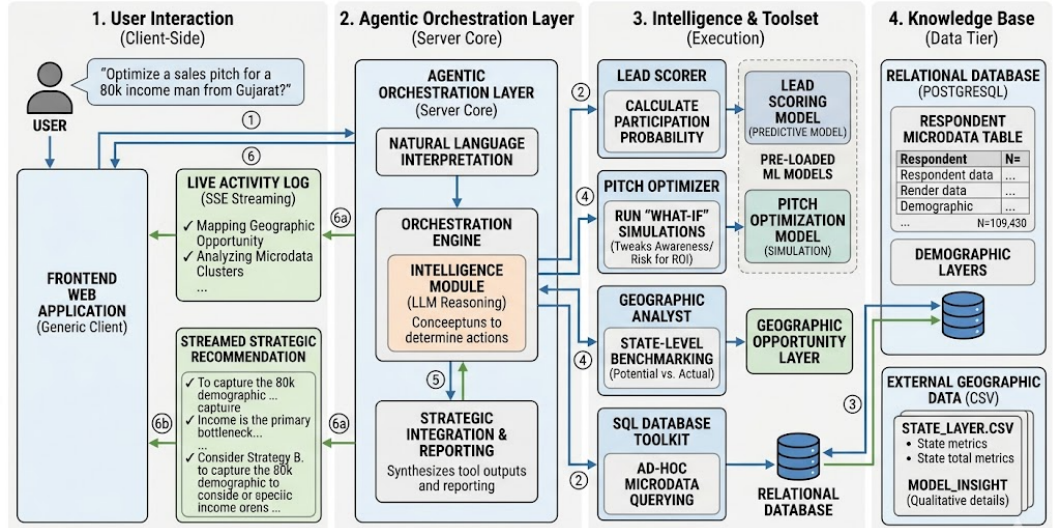
E. Regulate financial influencers

Social media advice is linked to speculative behaviour. Require qualification disclosure and ban unverified return claims.

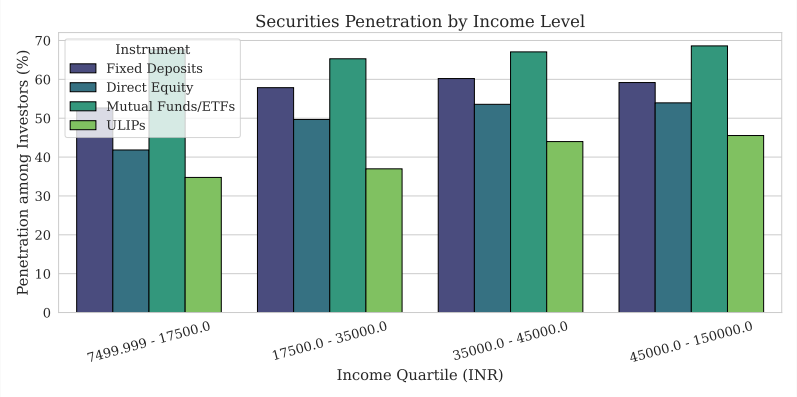
F. Test knowledge before derivatives

Many investors overestimate their understanding and end up in high-risk products. Replace self-declaration with a short knowledge test.

Implementation: Stockbroker Sales AI

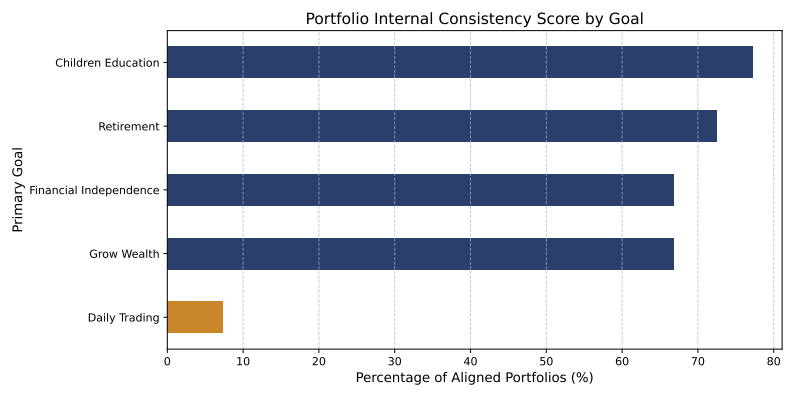


Appendix: Holdings by Income Quartile



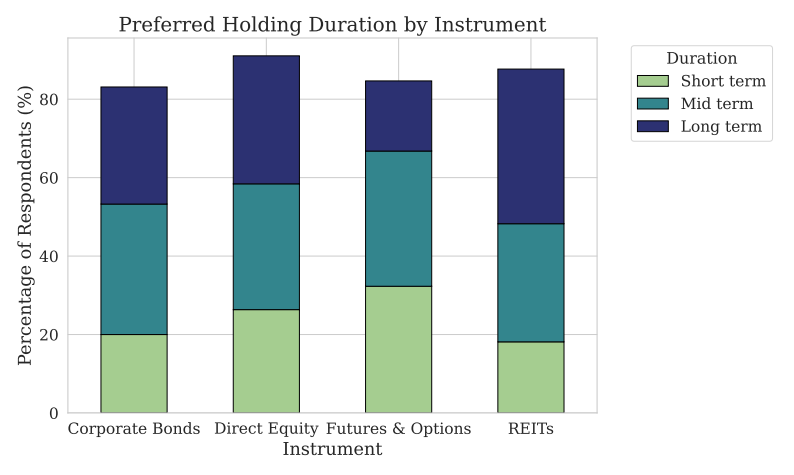
Market-linked instruments scale sharply with wealth.

Appendix: Goal-Portfolio Consistency



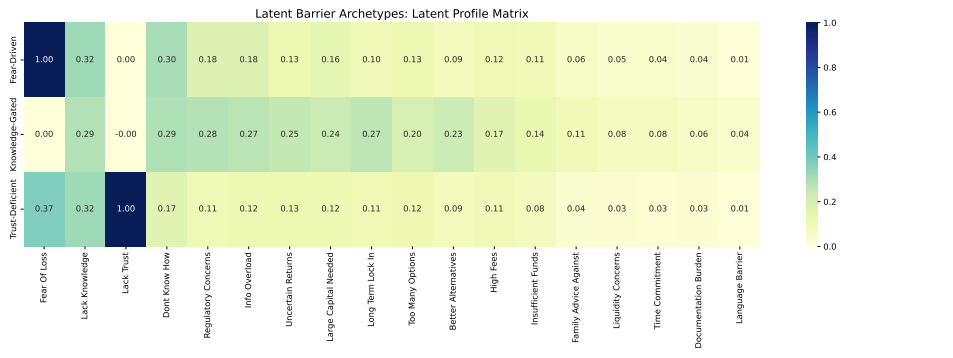
Consistency score by stated primary goal. Daily Trading cohort: 7.3%.

Appendix: Investment Horizons by Asset Class



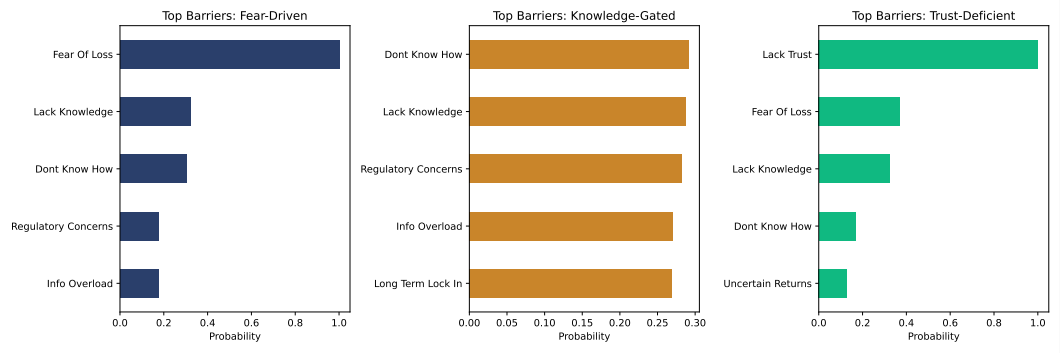
Equities and MFs held long-term; derivatives skew short-term.

Appendix: Barrier Archetype Profiles (Centroids)



Centroid matrix of the 18-dimensional barrier space. Values represent the probability of each barrier being present in the archetype profile.

Appendix: Archetype Defining Peaks



Top 5 barriers for each archetype. These peaks define the strategic naming: Fear-Driven (Fear of Loss), Knowledge-Gated (Lack Knowledge), and Trust-Deficient (Lack Trust).

Appendix: Full Logistic Regression Table

Predictor	**OR**	**95% CI**	z	Sig.
Product Awareness	2.218	[2.166, 2.272]	65.2	***
Log(Income)	1.603	[1.564, 1.643]	37.4	***
Education (yrs)	1.556	[1.518, 1.594]	35.6	***
Male	1.526	[1.490, 1.564]	33.9	***
Urban	1.455	[1.422, 1.489]	32.0	***
Risk Tolerance	1.170	[1.145, 1.196]	14.0	***
Millennial	1.124	[1.098, 1.151]	9.8	***
Gen X	0.993	[0.969, 1.017]	−0.6	
Baby Boomer	0.907	[0.881, 0.934]	−6.6	***
North	0.850	[0.825, 0.876]	−10.7	***
South	0.842	[0.816, 0.869]	−10.7	***
West	1.021	[0.989, 1.054]	1.3	

$N = 97{,}626$ McFadden $R^2 = 0.290$ *** $p < 0.001$, HC3 robust SE